



Daffodil International University

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Signature of Invigilator

Department/Program... Software Engineering

Examination : ☒ Midterm / ☒ Semester Final

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Course Title : Introduction to Software Engineering

Course Code : SE-113

Section : 46-B Date : 06. 11. 2025

Read the instructions carefully and follow them strictly:

1. Every student must write his/her ID number, Section, Semester, Course Title and Course Code clearly in the respective place.
2. Every student may use additional answer sheets, if needed, but he/She must write his/her ID number and other particulars clearly and get them attached with the original answer script duly signed by the invigilator.
3. Students, if found with any objectionable paper under their possession, shall be debarred from appearing in the examination.
4. Students must Maintain Silence and follow strict discipline in the examination Hall.
5. Cell phone is strictly prohibited in the examination Hall.
6. No student will be allowed to enter the examination Hall half an hour after the examination has started.
7. Students must sign in the attendance sheet before leaving the examination Hall.
8. Students will not be allowed to go outside the examination Hall during the examination.
9. Students must not write anything on the question paper, desk, chair, walls of the examination Hall.
10. For any matter pertaining to smooth conduct of examination not mentioned above, the decision of the invigilator/Hall in-charge shall be considered final.

Question No.	Full Marks	Marks Obtained
1.	18	17
2.	07	07
3.		
4.		
5.		
6.		
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9.		
10.		
11.		
12.		
13.		
14.		
15.		
Total	25	24

[Signature]

Signature of Examiner

Ans. to the Q. NO-1(a)

The traditional model of software development is waterfall model. Waterfall model is more suitable for projects which are stable and Agile software development methodology is suitable for projects which are changing requirements.

Waterfall (Traditional)	Agile
i) Waterfall is a linear and sequential model where requirements are fixed and each phase must be completed before the next phase begins.	ii) Agile is a software framework where projects are divided into sprints, iteration and continuous updated the requirements.
ii) Requirements are fixed. It is not a dynamic model.	ii) Requirements changing frequently.
iii) Easier to use and development is stable.	iii) Needed experienced programmers.
iv) EAC Testing and developing must be done during the processing phase.	iv) Testing and developing is done in every iterations.

if requirements are stable, fixed, we can use waterfall model and changing requirements, fast delivery we should use Agile methodology.

Ans. to the Q. NO-1(b)

Yes, Rafi's activities violate the intellectual property and uses anyone computer, ~~unethical~~ hack, professional responsibilities of a software engineer.

As a software engineer, we should follow many ethics. We have many professional responsibilities. These are -

i) Intellectual property - Government have many laws to protect the intellectual property of IT background people. Such as - patent, copyright etc. But in this scenario, Rafi copies a paid version of a popular learning software from

the internet and installs it - which are completely a crime. We can't use any paid activities of anyone.

ii) uses anyone computer: we can't misuse our competence or capability and use other computers for any forbidden works. Here, Rafi's do this.

iii) unethical hacking - Rafi uses his company's computer during office hours to hack into a competitor's server to check their product design - which is crime because as a software engineer, we shouldn't misuse our capability. We have no right to unethical hacking.

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Ans. to the Q. NO-1(c)

As a software engineer, our goal is to develop high-quality software. The quality attributes that should be considered in the development process are given below -

i) Reliability - the software should be finished at for a defined time, meeting the user requirements.

ii) Usability - The software should be easy for the clients to use, control and communicate the software.

iii) Efficiency - The software uses system resources like - CPU, memory etc. as a good manner.

iv) Testability - The software have not major flaws and errors.

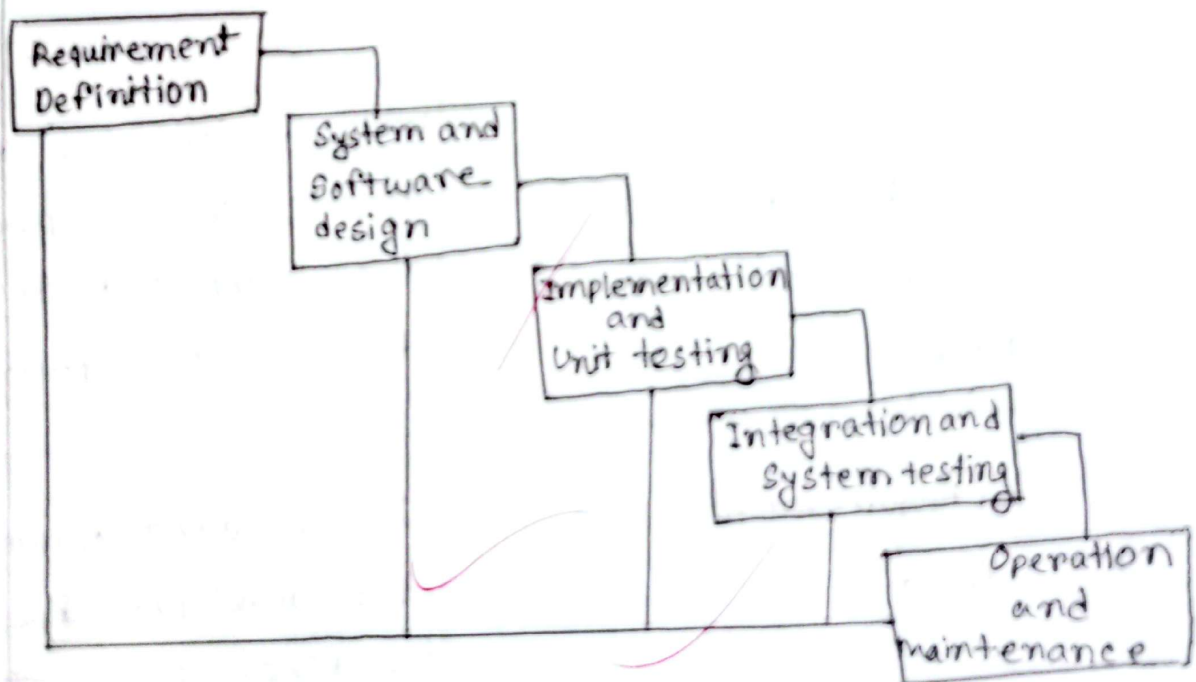
v) Portability - The software can use many different types of hardware.

vi) Maintainability - Easy to maintain, add new features, without changing big systems.

vii) Flexibility - The should add not face major problem to adopt new environment.

Ans. to the Q. NO- 1(d)

Waterfall model is a traditional model. It is a sequential and linear model in which each phase must be completed the next phase begins. The waterfall model's phases are-



Here, elaborating the advantages and many disadvantages of using the waterfall model for any software project in the software development-

Advantages	Disadvantages
i) Waterfall model is suitable for small projects which requirements aren't frequently changed.	i) It is desirable for easy and stable projects.
ii) Each phase must be completed before the next begins.	ii) Errors can't be identified easily.
iii) It is easy to maintain for its linear and sequential flow.	iii) Small changes can create major problems in the completed development process or software.
iv) Requirements are fixed.	iv) Users can't communicate the developers from before the final product is introduced.

Ans. to the Q.NO-2(a)

Based on the scenario, a The Spiral or risk-driven model is suitable software development life cycle (SDLC) for this project.

Spiral model is such type of software development model where the development process is done by many cycles which are completed. It is a combination of risk analysis and solving. The spiral model is used for complex, major projects to solve the real-world problems. Here the diagram is of spiral model -

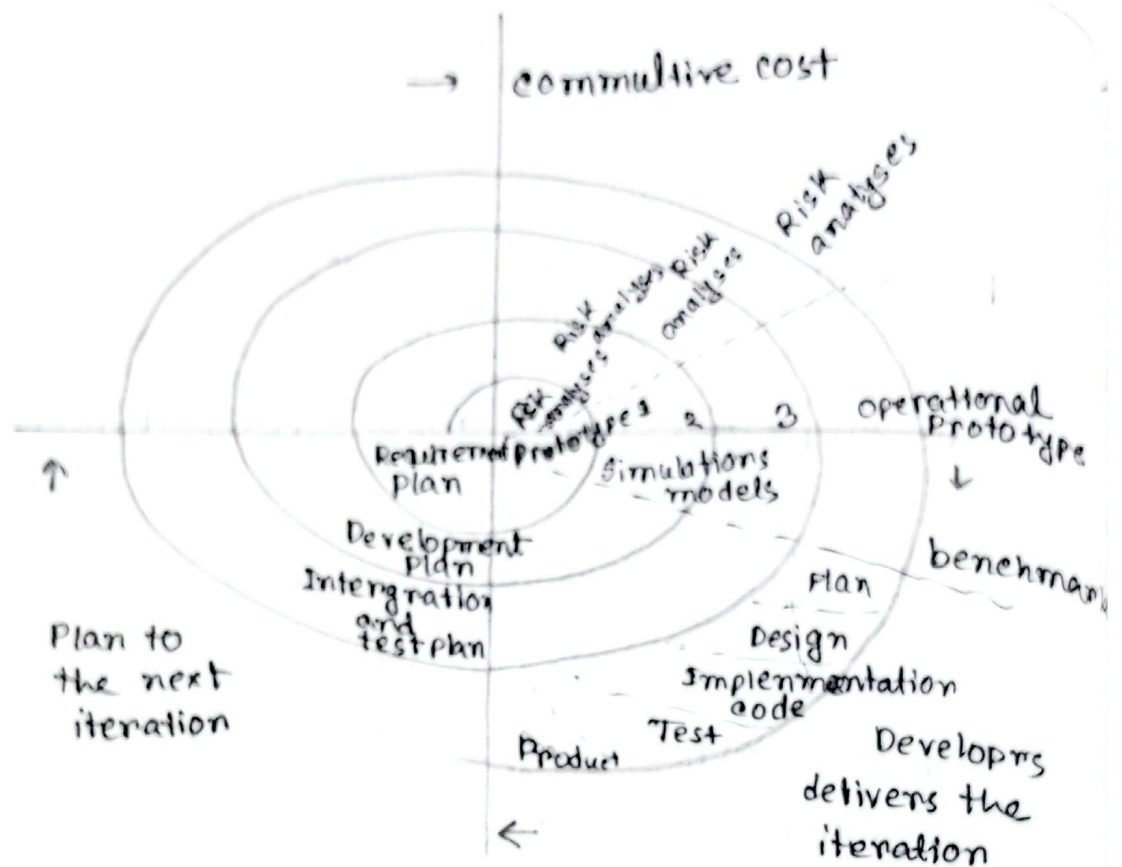


Diagram of Spiral model

In this scenario, the software collects real-time data, the project have high technical and financial risks, requirements are not fully defined and client also see the early prototype before the final system design, — all of these indicates the spiral model is appropriate for this software. The spiral model works four functions —

- i) Planning
- ii) Risk analysis and design
- iii) Develop and Implementation

m) Evaluation

In a Spiral model, requirements are not well-defined at the starting, users can get feel the actual system by use for using prototype (demo version) of the software, in every cycle risk are analyzed and solved the user's & feedback throughout the development process. to ensure that the system can adapt to real-world conditions. In this model, users can easily provide feedback for this reason the system is more reliable, ~~main~~ not failure. Spiral model

Ans. to the Q. NO-2(b)

is suitable for complex project and long-term project. Here, requirements are updated based on the stakeholders requirements on real-life problems.

In conclude, based on the scenario the Spiral model is ~~no~~ appropriate for this real-world project.

Ans. to the Q.NO-2(b)

According to the scenario, the Spiral model is suitable for this project. Demonstrating each phase that specifically addresses the needs and challenges of the SDLC (Software development life cycle) - problems -

Needs

- i) Spiral model is used for complex and long term project to solve real-world problems.
- ii) The development processes are more accurate because risk analysis is happened highly.

iii)

challenges

- i) By using spiral model, sometimes spirals go to indefinitely.
- ii) Sometimes, we need to be finished the project immediately which can create pressure to the developers, and the project should.
- iii) Misinformation or misnot proper communication with between user and developers arise the risk of lives.
- iv) The cost and of the ~~fa~~ software failure can be harmful for the lives means lives are at risk.
- v) The development process is time-consuming sometimes.
- vi) In turnaround time, requirement changes are changed frequent quickly.